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← Modules

Motors & Actuators & Drivers

SimpleFOCmini FOC Brushless DC Motor Driver Board (SKU: DRI0058)

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DC Geared Motor with Encoder (SKU: FIT0186)

TT Motor with Encoder (6V 160RPM 120:1) (SKU: FIT0450)

Gravity: Digital 5A Relay Module (SKU: DFR0017)

Gravity: Digital Peristaltic Pump Module (SKU: DFR0523)

micro:Driver – Driver Expansion Board for micro:bit / UNIHIKER M10 / K10 (SKU: DFR0548)

Gravity: Digital 10A Relay Module (SKU: DFR0473)

Dual Channel 12A DC Motor Driver Board (SKU: DFR0601)

DSS M15S 270° Metal Servo Module (SKU: SER0044)

M0601 Integrated Hub Motor (SKU: FIT1042)

Brushless DC Motor with Encoder 12V 159RPM (SKU: FIT0441)

12V Metal DC Geared Motor with Encoder (90:1, 122RPM, 38Kg.cm) (SKU: FIT0403)

2x7A DC Motor Driver Board (SKU: DRI0041)

Gravity: TM6605 Haptic Motor Driver Module (SKU: DRI0056)

Gravity: Dual Bipolar Stepper Motor Shield for Arduino (DRV8825) (SKU: DRI0023)

wiki / Modules / Motors & Actuators & Drivers / DRI0058

SimpleFOCmini FOC Brushless DC Motor Driver Board



SKU: DRI0058

The SimpleFOCmini brushless DC motor driver board simply connects to the control signal and the motor's UVW three-phase wires to achieve open/closed-loop control of the position, speed, and torque of a BLDC motor. It supports SPWM and SVPWM control algorithms, but not the traditional 6-step commutation control algorithm. Using it with a three-phase brushless DC motor allows for quick startup using the SimpleFOC open-source library, and is compatible with development boards such as Arduino, ESP32, and Raspberry Pi.

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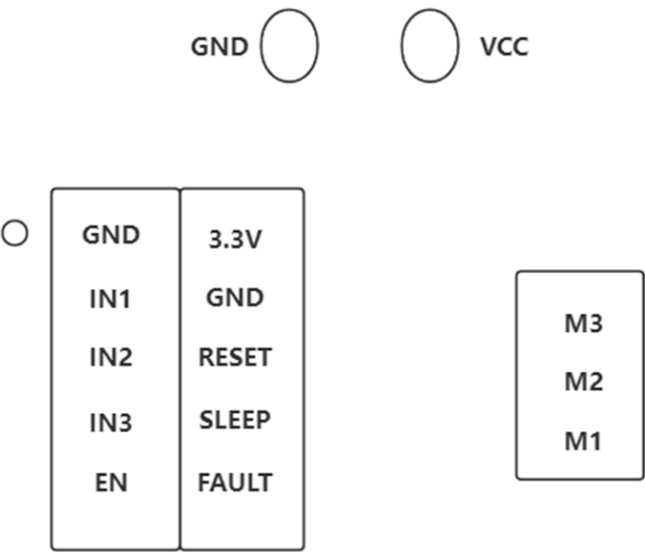
Dimension Schematics STP Files

Docs Tech Specs

Specification

Parameter	Value
Input power supply voltage	8–30V
Maximum output current per channel	2.5A
Onboard 3.3L LDO output	Maximum current 10mA, maximum PWM frequency 200kHz
Control signal compatibility	3.3V/5V
Dimensions	26*21.5mm

Pinout



Pin	Function	Description
VCC	Positive input power	Input power 8~30V
GND	Negative input power	
IN1	Channel 1 input	When IN1 is high, M1 is also high; when IN1 is low, M1 is also low
IN2	Channel 1 input	When IN2 is high, M2 is also high; when IN2 is low, M2 is also low
IN3	Channel 1 input	When IN3 is high, M3 is also high; when IN3 is low, M3 is also low
EN	Enable input	Simultaneously controls the enable of Channels 1, 2, and 3. Input is valid when high
3.3V	3.3V power supply	3.3V power supply output, maximum output current 10mA
RESET	Reset input	Sets the chip to summer position and disables the output. Active low
SLEEP	Low-power mode	Low-power sleep mode, active low
FAULT	Fault signal output	Outputs protection signals for overtemperature and overcurrent, active low
M1	Output channel 1	Output terminal, follows the IN1 signal, and the output amplitude is the power supply voltage
M2	Output channel 2	Output terminal, follows the IN2 signal, and the output amplitude is the power supply voltage
M3	Output channel 3	Output terminal, follows the IN3 signal, and the output amplitude is the power supply voltage

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